EVERY SUBSET OF A COUNTABLE-APPROXIMATION TOPOLOGICAL RAMSEY SPACE IS RAMSEY IN THE SOLOVAY MODEL

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Editorial note. Complete original author definitions, supporting proofs and final correction remark retained; one primary per task. No proof repair, added assumptions or mathematical correctness certification.

Editorial note. Mechanical preamble adapter blanks only native biblatex and addbibresource setup; installed BibTeX uses original references.bib. Native printbibliography line is omitted from body spans and bibliography is invoked by the assembler.

Allowed standard prerequisites (editorial).

- Set-theoretic forcing and topological Ramsey spaces. Complete native axioms, acceptance/rejection lemmas, capture definition and cited Solovay collapse facts retained.. The selected genericity, pure-decision or collapse argument remains present as counted author proof.

1. Introduction

Let $[\mathbb{N}]^\infty$ denote the set of infinite subsets of the set of natural numbers $\mathbb{N}$. Given an infinite $A \subseteq \mathbb{N}$ and a finite $a \subseteq A$, $[a, A]$ denotes the set of all infinite subsets $B \subseteq A$ such that a is an initial segment of B (i.e. $B \cap (\max(a) + 1) = a$, where $\max(\emptyset) := -1$). A subset $\mathcal{X} \subseteq [\mathbb{N}]^\infty$ is *(completely) Ramsey* if for all $A \in [\mathbb{N}]^\infty$ and finite $a \subseteq A$, there exists some $B \in [a, A]$ such that $[a, B] \subseteq \mathcal{X}$ or $[a, B] \subseteq \mathcal{X}^c$.

Theorem 1.1 (Mathias, [3])**.** *Let κ be an inaccessible cardinal, and let G be $\mathrm{Coll}(\omega, <\kappa)$-generic. Then in $\mathbf{L}(\mathbb{R})^{\mathbf{V}[G]}$, every subset of $[\mathbb{N}]^\infty$ is Ramsey.*

Much of the classical Ramsey theory of $[\mathbb{N}]^\infty$ has been extended to topological Ramsey spaces (i.e. closed triples $(\mathcal{R}, \leq, r)$ satisfying the axioms **A1**–**A4** introduced by Todorčević in [5]), so it is natural to ask whether Theorem 1.1 holds in this generality. The key obstacle lies in Mathias' use of selective ultrafilters in his arguments — while the almost subset relation $\subseteq^*$ serves as a natural σ-closed quasi-order on $[\mathbb{N}]^\infty$ that allows diagonalisation of sequences, there is no analogue that is σ-closed for topological Ramsey spaces in general. In [2], Di Prisco–Mijares–Nieto developed the abstract Mathias forcing $\mathbb{M}_\mathcal{R}$, and proved that $\mathbb{M}_\mathcal{R}$ satisfies the Mathias and Prikry properties if the almost reduction relation $\leq^*$ (where $A \leq^* B$ iff $\emptyset \neq [a, A] \subseteq [a, B]$ for some $a \in \mathcal{AR}$) is σ-closed. While the σ-closed hypothesis on $\leq^*$ was removed in the author's paper [6] (so $\mathbb{M}_\mathcal{R}$ satisfies both the Mathias and Prikry properties for every topological Ramsey space), a proof of the corresponding generalisation of Theorem 1.1 has not appeared in the literature.

In this paper, we review the axioms of topological Ramsey spaces and the abstract Mathias forcing, and prove the following:

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Theorem 1.2. *Let $(\mathcal{R}, \leq, r)$ be a closed triple satisfying **A1**–**A4** such that $\mathcal{AR}$ is countable. Let κ be an inaccessible cardinal, and let G be $\mathrm{Coll}(\omega, <\kappa)$-generic. Then in $\mathbf{L}(\mathbb{R})^{\mathbf{V}[G]}$, every subset of $\mathcal{R}$ is Ramsey.*

2. Topological Ramsey spaces

The axiomatisation of topological Ramsey spaces was initially studied by Carlson and Simpson in [1]. In [5], Todorčević presented the four axioms, labelled **A1**–**A4**, which are sufficient for a closed triple $(\mathcal{R}, \leq, r)$ to be a topological Ramsey space. We shall recap the axioms in this section.

Here, $\mathcal{R}$ is a non-empty set, $\leq$ is a quasi-order on $\mathcal{R}$, and $r : \mathcal{R} \times \omega \to \mathcal{AR}$ is a surjective function. We also define a sequence of maps $r_n : \mathcal{R} \to \mathcal{AR}$ by $r_n(A) := r(A, n)$ for all $A \in \mathcal{R}$. Let $\mathcal{AR}_n \subseteq \mathcal{AR}$ be the image of r_n (i.e. $a \in \mathcal{AR}_n$ iff $a = r_n(A)$ for some $A \in \mathcal{R}$).

Axiom (**A1**, Sequencing)**.**

(1) $r_0(A) = \emptyset$ for all $A \in \mathcal{R}$.
(2) $A \neq B$ implies $r_n(A) \neq r_n(B)$ for some n.
(3) $r_n(A) = r_m(B)$ implies $n = m$ and $r_k(A) = r_k(B)$ for all $k < n$.

For each $a \in \mathcal{AR}$, let $\mathrm{lh}(a)$ denote the unique n such that $a \in \mathcal{AR}_n$. By Axiom **A1**(3), this n is well-defined. Axiom **A1**(3) also implies that if $a \in \mathcal{AR}_n$ and $k \leq n$, then $r_k(A) = r_k(B)$ for any $A, B \in \mathcal{R}$ with $r_n(A) = r_n(B) = a$, so we may unambiguously extend r_k to $\{a \in \mathcal{AR} : \mathrm{lh}(a) \geq k\}$ by letting $r_k(a)$ denote this common value. For $a, b \in \mathcal{AR}$, we write $a \sqsubseteq b$ iff $a = r_{\mathrm{lh}(a)}(b)$, and for $B \in \mathcal{R}$, we write $a \sqsubseteq B$ iff $a = r_{\mathrm{lh}(a)}(B)$.

Axiom (**A2**, Finitisation)**.** There is a quasi-ordering $\leq_{\mathrm{fin}}$ on $\mathcal{AR}$ such that:

(1) $\{a \in \mathcal{AR} : a \leq_{\mathrm{fin}} b\}$ is finite for all $b \in \mathcal{AR}$.
(2) $B \leq A$ iff $\forall n \exists m [r_n(B) \leq_{\mathrm{fin}} r_m(A)]$.
(3) $\forall a, b, c \in \mathcal{AR} [a \sqsubseteq b \wedge b \leq_{\mathrm{fin}} c \to \exists d \sqsubseteq c [a \leq_{\mathrm{fin}} d]]$.

We may define the *Ellentuck neighbourhoods* as follows: For any $A \in \mathcal{R}$, $a \in \mathcal{AR}$ and $n \in \mathbb{N}$, we let:

$$\begin{aligned} [a, A] &:= \{B \in \mathcal{R} : a \sqsubseteq B \wedge B \leq A\}, \\ [n, A] &:= [r_n(A), A], \\ [A] &:= [\emptyset, A], \\ [a] &:= \{B \in \mathcal{R} : a \sqsubseteq B\}. \end{aligned}$$

We also define the depth function depth_A, for $A \in \mathcal{R}$ and $a \in \mathcal{AR}$, by:

$$\mathrm{depth}_A(a) := \begin{cases} \min\{n < \omega : a \leq_{\mathrm{fin}} r_n(A)\}, & \text{if such } n \text{ exists}, \\ \infty, & \text{otherwise.} \end{cases}$$

We also define, for each $A \in \mathcal{R}$ and $n < \omega$:

$$\begin{aligned} \mathcal{AR}{\restriction}A &:= \{a \in \mathcal{AR} : \exists m [a \leq_{\mathrm{fin}} r_m(A)]\}, \\ \mathcal{AR}_n{\restriction}A &:= \mathcal{AR}_n \cap \mathcal{AR}{\restriction}A. \end{aligned}$$

If $a \in \mathcal{AR}{\restriction}A$, then:

$$\begin{aligned}
\mathcal{AR}{\restriction}[a, A] &:= \{b \in \mathcal{AR}{\restriction}A : a \sqsubseteq b\},\\
\mathcal{AR}{\restriction}[a] &:= \{b \in \mathcal{AR} : a \sqsubseteq b\},\\
\mathcal{AR}{\restriction}a &:= \{b \in \mathcal{AR} : b \leq_{\text{fin}} a\},\\
\mathcal{AR}_n{\restriction}[a, A] &:= \mathcal{AR}_n \cap \mathcal{AR}{\restriction}[a, A],\\
\mathcal{AR}_n{\restriction}[a] &:= \mathcal{AR}_n \cap \mathcal{AR}{\restriction}[a],\\
\mathcal{AR}_n{\restriction}a &:= \mathcal{AR}_n \cap \mathcal{AR}{\restriction}a,\\
r_n[a, A] &:= \{b \in \mathcal{AR}{\restriction}[a, A] : \text{lh}(b) = n\},\\
r_n[A] &:= r_n[\emptyset, A].
\end{aligned}$$

Note that for all $a \in \mathcal{AR}$, $\text{depth}_A(a) < \infty$ iff $a \in \mathcal{AR}{\restriction}A$.

Axiom (**A3**, Amalgamation)**.** Let $A \in \mathcal{R}$.

(1) For all $a \in \mathcal{AR}{\restriction}A$ and $B \in [\text{depth}_A(a), A]$, $[a, B] \neq \emptyset$.
(2) If $B \leq A$ and $a \in \mathcal{AR}{\restriction}B$, then there exists $B' \in [\text{depth}_A(a), A]$ such that $\emptyset \neq [a, B'] \subseteq [a, B]$.

Axiom (**A4**, Pigeonhole)**.** For all $A \in \mathcal{R}$, $a \in \mathcal{AR}{\restriction}A$ and $\mathcal{O} \subseteq \mathcal{AR}_{\text{lh}(a)+1}$, there exists some $B \in [\text{depth}_A(a), A]$ such that $r_{\text{lh}(a)+1}[a, B] \subseteq \mathcal{O}$ or $r_{\text{lh}(a)+1}[a, B] \subseteq \mathcal{O}^c$.

Note that, by **A3**, the B in **A4** may instead be taken to be an element of $[a, A]$. By Axiom **A1**, we may identify each element $A \in \mathcal{R}$ with a sequence of elements of $\mathcal{AR}$, via the map $A \mapsto (r_n(A))_{n<\omega}$. Therefore, we may identify $\mathcal{R}$ with a subset of $\mathcal{AR}^{\mathbb{N}}$.

We shall focus on the following topologies on $\mathcal{R}$.

- The *metric topology* induced by the first difference metric, where for $A, B \in \mathcal{R}$, $d(A, A) = 0$ and $d(A, B) = \frac{1}{2^n}$ for $A \neq B$, where n is the least integer such that $r_n(A) \neq r_n(B)$. The sets of the form $[a]$ for $a \in \mathcal{AR}$ form a neighbourhood base of this topology. If $\mathcal{AR}$ is countable, then $\mathcal{R}$ is a Polish space under this metric topology.
- The *Ellentuck topology* generated by open sets of the form $[a, A]$ for $A \in \mathcal{R}$ and $a \in \mathcal{AR}{\restriction}A$.

Unless stated otherwise, all topological properties of subsets of $\mathcal{R}$ mentioned in this paper are with respect to the metric topology.

Definition 2.1. We say that $(\mathcal{R}, \leq, r)$ is a *closed triple* if $\mathcal{R}$ is a metrically closed subset of $\mathcal{AR}^{\mathbb{N}}$.

Definition 2.2. Let $(\mathcal{R}, \leq, r)$ be a closed triple satisfying **A1** and **A2**. A subset $\mathcal{X} \subseteq \mathcal{R}$ is *Ramsey* if for all $A \in \mathcal{R}$ and $a \in \mathcal{AR}{\restriction}A$, there exists some $B \in [a, A]$ such that $[a, B] \subseteq \mathcal{X}$ or $[a, B] \subseteq \mathcal{X}^c$. If $[a, B] \subseteq \mathcal{X}^c$ always holds, then $\mathcal{X}$ is *Ramsey null*.

Todorčević subsequently showed in [5] the following theorem:

Theorem 2.3 ([5, Abstract Ellentuck theorem])**.** *Let $(\mathcal{R}, \leq, r)$ be a closed triple satisfying **A1**–**A4**. Then a subset of $\mathcal{R}$ is Ramsey iff it has the property of Baire under the Ellentuck topology, and it is Ramsey null iff it is meagre under the Ellentuck topology.*

In particular, if $\mathcal{AR}$ is countable (so the metric topology is Polish), then every analytic set is Ramsey. A closed triple $(\mathcal{R}, \leq, r)$ satisfying **A1** and **A2** is a *topological Ramsey space* if it satisfies the conclusion of Theorem 2.3. Theorem 2.3 thus asserts that every closed triple satisfying **A1**–**A4** is a topological Ramsey space.

We shall prove a lemma asserting that the axioms **A1**–**A3** allow us to perform diagonalisation over a family of dense open sets. The following definition is a variant of [2, Definition 3.6], and was introduced earlier by the author in [7, 6].

Definition 2.4. Let $A \in \mathcal{R}$ and $a \in \mathcal{AR}{\restriction}A$.

- A family of subsets $\vec{\mathcal{D}} = \{\mathcal{D}_b\}_{b \in \mathcal{AR}{\restriction}[a,A]}$ is *dense open below* $[a, A]$ if for all $b \in \mathcal{AR}{\restriction}[a, A]$, $\mathcal{D}_b$ is a $\leq$-downward closed subset of $[b, A]$, and for all $B \in [b, A]$, there exists some $C \in [b, B]$ such that $C \in \mathcal{D}_b$.
- Let $\vec{\mathcal{D}} = \{\mathcal{D}_b\}_{b \in \mathcal{AR}{\restriction}[a,A]}$ be dense open below $[a, A]$. We say that $B \in [a, A]$ *diagonalises* $\vec{\mathcal{D}}$ if for all $b \in \mathcal{AR}{\restriction}[a, B]$, there exists some $A_b \in \mathcal{D}_b$ such that $[b, B] \subseteq [b, A_b]$.

Lemma 2.5. *Let $(\mathcal{R}, \leq, r)$ be a closed triple satisfying **A1**–**A3**, and let $A \in \mathcal{R}$. Let $\vec{\mathcal{D}} = \{\mathcal{D}_b\}_{b \in \mathcal{AR}{\restriction}A}$ be a family of subsets that is dense open below $[\emptyset, A]$. Then for all $a \in \mathcal{AR}{\restriction}A$, there exists some $B \in [a, A]$ such that for all $b \in \mathcal{AR}{\restriction}B$ with $\mathrm{depth}_B(b) \geq \mathrm{lh}(a)$, there exists some $A_b \in \mathcal{D}_b$ such that $[b, B] \subseteq [b, A_b]$.*

Proof. Fix some $A \in \mathcal{R}$ and $a \in \mathcal{AR}{\restriction}A$, and suppose that $\vec{\mathcal{D}} = \{\mathcal{D}_b\}_{b \in \mathcal{AR}{\restriction}A}$ is dense open below $[\emptyset, A]$. We shall define a fusion sequence $(A_n)_{n<\omega}$ in $[a, A]$, with $a_n = r_{\mathrm{lh}(a)+n}(A_n)$, such that $A_{n+1} \in [a_n, A_n]$: By **A3**(1), let $A_0 \in [a, A]$, and suppose that A_n has been defined. Let $\{b_n^i : i < N\}$ enumerate the set of all $b \in \mathcal{AR}{\restriction}A_n$ such that $\mathrm{depth}_{A_n}(b) = \mathrm{lh}(a) + n$, which is finite by **A2**(1). We define another sequence $(A_n^i)_{i \leq N}$ in $[a_n, A_n]$ as follows: Let $A_n^0 := A_n$. Suppose that $A_n^i \in [a_n, A_n]$ has been defined. Then $r_j(A_n^i) = r_j(A_n)$ for all $j \leq \mathrm{lh}(a) + n$, so $\mathrm{depth}_{A_n^i}(b_n^i) = \mathrm{depth}_{A_n}(b_n^i) = \mathrm{lh}(a) + n$, and by **A3**(1) we may let $B_n^i \in [b_n^i, A_n^i]$. Since $\mathcal{D}_{b_n^i}$ is dense open in $[b_n^i, A]$, there exists some $A_{b_n^i} \in \mathcal{D}_{b_n^i}$ and $C_n^i \in [b_n^i, B_n^i]$ such that $[b_n^i, C_n^i] \subseteq [b_n^i, A_{b_n^i}]$. By **A3**(2), we may let $A_n^{i+1} \in [\mathrm{depth}_{A_n^i}(b_n^i), A_n^i] = [a_n, A_n^i]$ be such that $[b_n^i, A_n^{i+1}] \subseteq [b_n^i, C_n^i]$. We complete the induction by letting $A_{n+1} := A_n^N$. Since $\mathcal{R}$ is closed, $B := \lim_{n\to\infty} a_n$ is an element of $\mathcal{R}$, and $B \in [a, A]$ and $B \leq A_n^i$ for all n and $i \leq N$ by **A2**(2). We shall show that B is the required element of $[a, A]$.

Let $b \in \mathcal{AR}{\restriction}B$ be such that $\mathrm{depth}_B(b) = \mathrm{lh}(a)+n$ for some $n \geq 0$. Since $r_j(B) = r_j(A_n)$ for all $j \leq \mathrm{lh}(a) + n$, we have that $\mathrm{depth}_{A_n}(b) = \mathrm{lh}(a) + n$, so $b = b_n^i$ for some i in the induction step of the proof. Then $[b, B] \subseteq [b, A_n^{i+1}] \subseteq [b, C_n^i] \subseteq [b, A_b]$ and $A_b \in \mathcal{D}_b$, as desired. □

In particular (as $\mathrm{depth}_B(b) \geq \mathrm{depth}_B(a) = \mathrm{lh}(a)$ for all $b \in \mathcal{AR}{\restriction}[a, B]$), we have the following:

Lemma 2.6 ([7, Lemma 3.16])**.** *Let $(\mathcal{R}, \leq, r)$ be a closed triple satisfying **A1**–**A3**, and let $A \in \mathcal{R}$ and $a \in \mathcal{AR}{\restriction}A$. Then every family of subsets $\vec{\mathcal{D}} = \{\mathcal{D}_b\}_{b \in \mathcal{AR}{\restriction}[a,A]}$ that is dense open below $[a, A]$ has a diagonalisation.*

3. Abstract Mathias forcing

We recap the abstract Mathias forcing introduced in [2], and prove that it has the Mathias and Prikry properties. Most of the content in this section is adapted from [6, §6].

Fix a closed triple $(\mathcal{R}, \leq, r)$ satisfying **A1**–**A4**. The abstract Mathias forcing is defined as the forcing poset $(\mathbb{M}_{\mathcal{R}}, \leq)$, where:

$$\mathbb{M}_{\mathcal{R}} = \{(a, A) : A \in \mathcal{R} \text{ and } a \in \mathcal{AR}{\restriction}A\},$$

and that $(b, B) \leq (a, A)$ iff $[b, B] \subseteq [a, A]$. We also write $(b, B) \leq_0 (a, A)$ if $(b, B) \leq (a, A)$ and $b = a$.

Definition 3.1. An element $A \in \mathcal{R}$ is said to be $\mathbb{M}_{\mathcal{R}}$-generic (over a ground model $\mathbf{V}$) if:

$$G_A := \{(a, B) \in \mathbb{M}_{\mathcal{R}} : A \in [a, B]\}$$

is a generic filter of $\mathbb{M}_{\mathcal{R}}$ (over $\mathbf{V}$).

Lemma 3.2.

(1) If G is a generic filter of $\mathbb{M}_{\mathcal{R}}$, then there exists a unique $\mathbb{M}_{\mathcal{R}}$-generic $A_G \in \mathcal{R}$ such that $G = \{(a, A) \in \mathbb{M}_{\mathcal{R}} : A_G \in [a, A]\}$.
(2) If $A \in \mathcal{R}$ is $\mathbb{M}_{\mathcal{R}}$-generic, then $A_{G_A} = A$.
(3) If G is a generic filter of $\mathbb{M}_{\mathcal{R}}$, then $G_{A_G} = G$.

Proof. We only prove (1) — (2) follows from the uniqueness of A_G in (1), and (3) follows easily from the definitions. Let $\mathcal{D}_n := \{(a, A) \in \mathbb{M}_{\mathcal{R}} : \mathrm{lh}(a) \geq n\}$. We see that $\mathcal{D}_n$ is dense open in $\mathbb{M}_{\mathcal{R}}$ for all n — if $(a, A) \notin \mathcal{D}_n$ (i.e. $\mathrm{lh}(a) < n$) and $b \in r_n[a, A]$, then by **A3**(1) we have that $[b, A] \neq \emptyset$. Let $B \in [b, A]$, and we have that $[b, B] \subseteq [a, A]$, so $(b, B) \leq (a, A)$ and $(b, B) \in \mathcal{D}_n$. Therefore, $G \cap \mathcal{D}_n \neq \emptyset$ for all n, and since G is a filter, for all $(a, A), (b, B) \in G$, $a \sqsubseteq b$ or $b \sqsubseteq a$. Thus, there exists a unique $A_G \in \mathcal{R}$ such that for all n, $(r_n(A_G), A) \in G$ for some A. It's easy to verify that $G = G_{A_G}$. □

3.1. **Abstract combinatorial forcing.** We recap the abstract combinatorial forcing developed in [5, §4.3]. We fix a subset $\mathcal{X} \subseteq \mathcal{R}$.

Definition 3.3. Let $A \in \mathcal{R}$ and let $a \in \mathcal{AR}$. We say that:

(1) *A accepts a* if $[a, A] \subseteq \mathcal{X}$.
(2) *A rejects a* if $a \in \mathcal{AR}{\restriction}A$, and for all $B \in [a, A]$, B does not accept a.
(3) *A decides a* if A accepts or rejects a.

Lemma 3.4 ([5, Lemma 4.31])**.** *Let $A \in \mathcal{R}$ and let $a \in \mathcal{AR}$.*

(1) If $a \notin \mathcal{AR}{\restriction}A$, A accepts a (as $[a, A] = \emptyset$).
(2) If A accepts a and $B \leq A$, then B accepts a.
(3) If A rejects a, $B \leq A$ and $a \in \mathcal{AR}{\restriction}B$, then B rejects a.
(4) If A decides a and $B \leq A$, then B decides a.
(5) If $a \in \mathcal{AR}{\restriction}A$, then there exists some $B \in [\mathrm{depth}_A(a), A]$ that decides a.
(6) If $a \in \mathcal{AR}{\restriction}A$ and A decides a, then for all $B \in [\mathrm{depth}_A(a), A]$, B decides a in the same way A does.
(7) A accepts a iff A accepts every $b \in r_{\mathrm{lh}(a)+1}[a, A]$.

Lemma 3.5 ([5, Lemma 4.35])**.** *If A rejects a, then there exists some $B \in [a, A]$ which rejects every $b \in \mathcal{AR}{\restriction}[a, B]$.*

We refer readers to [5, §4] for the proofs.

3.2. **Mathias and Prikry properties.** We may now introduce the Mathias and Prikry properties.

Definition 3.6. Let $(\mathcal{R}, \leq, r)$ be a closed triple satisfying **A1**–**A4**.

(1) $\mathbb{M}_{\mathcal{R}}$ satisfies the *Mathias property* if whenever $A \in \mathcal{R}$ is $\mathbb{M}_{\mathcal{R}}$-generic over a ground model $\mathbf{V}$ and $B \leq A$, B is also $\mathbb{M}_{\mathcal{R}}$-generic over $\mathbf{V}$.
(2) $\mathbb{M}_{\mathcal{R}}$ satisfies the *Prikry property* if whenever φ is a sentence in the forcing language, then we have that for all $p \in \mathbb{M}_{\mathcal{R}}$, there exists some $q \leq_0 p$ such that $q \Vdash \varphi$ or $q \Vdash \neg\varphi$.

Given a subset $\mathcal{D} \subseteq \mathbb{M}_{\mathcal{R}}$, we let:

$$\overline{\mathcal{D}} := \bigcup_{(a,A)\in\mathcal{D}} [a, A].$$

Note that $\overline{\mathcal{D}}$ is an Ellentuck open subset of $\mathcal{R}$ in $\mathbf{V}$. Thus, if $\mathcal{D}$ is dense open in $\mathbb{M}_{\mathcal{R}}$, then for all $A \in \mathcal{R}$ and $a \in \mathcal{AR}{\restriction}A$, A does not reject a w.r.t. $\overline{\mathcal{D}}$ (otherwise, by Lemma 3.5 there exists some $B \in [a, A]$ that rejects every $b \in \mathcal{AR}{\restriction}[a, B]$, contradicting that $\mathcal{D}$ is dense in $\mathbb{M}_{\mathcal{R}}$). In other words, for all $A \in \mathcal{R}$ and $a \in \mathcal{AR}{\restriction}A$, there exists some $B \in [a, A]$ such that $[a, B] \subseteq \overline{\mathcal{D}}$.

Definition 3.7. Let $\mathcal{D} \subseteq \mathbb{M}_{\mathcal{R}}$. We say that $(a, A) \in \mathbb{M}_{\mathcal{R}}$ *captures* $\mathcal{D}$ if for all $B \in [a, A]$, there exists some initial segment $b \in \mathcal{AR}{\restriction}[a, B]$ of B such that $(b, A) \in \mathcal{D}$.

We remark that the statement "(a, A) captures $\mathcal{D}$" is absolute between transitive models: (a, A) captures $\mathcal{D}$ iff the tree $\{b \in \mathcal{AR}{\restriction}[a, A] : (c, A) \notin \mathcal{D}$ for all $c \sqsubseteq b$ with $a \sqsubseteq c\}$, ordered by $\sqsubseteq$, has no infinite branch, and well-foundedness is absolute between transitive models.

Lemma 3.8. *Let $\mathcal{D} \subseteq \mathbb{M}_{\mathcal{R}}$ be dense open. Then for all $A \in \mathcal{R}$ and $a \in \mathcal{AR}{\restriction}A$, there exists some $B \in [a, A]$ such that (a, B) captures $\mathcal{D}$.*

Proof. By the observation above, we may let $A' \in [a, A]$ be such that $[a, A'] \subseteq \overline{\mathcal{D}}$. For each $b \in \mathcal{AR}{\restriction}[a, A']$, let:

$$\mathcal{D}_b := \{C \in [b, A'] : (b, C) \in \mathcal{D} \text{ or } \forall C' \in [b, C]\, (b, C') \notin \mathcal{D}\}.$$

It's clear that $\{\mathcal{D}_b\}_{b \in \mathcal{AR}{\restriction}[a,A']}$ is dense open below $[a, A']$. By Lemma 2.6, there exists some $B \in [a, A']$ that diagonalises $\{\mathcal{D}_b\}_{b \in \mathcal{AR}{\restriction}[a,A']}$. We shall show that (a, B) captures $\mathcal{D}$.

Let $D \in [a, B]$. Since $D \in [a, A'] \subseteq \overline{\mathcal{D}}$, there exists some $(b, C) \in \mathcal{D}$ such that $D \in [b, C]$, and we may assume that $a \sqsubseteq b$. As B diagonalises $\{\mathcal{D}_b\}_{b \in \mathcal{AR}{\restriction}[a,A']}$ and $b \in \mathcal{AR}{\restriction}[a, B]$, there exists some $A_b \in \mathcal{D}_b$ such that $[b, B] \subseteq [b, A_b]$, so either $(b, A_b) \in \mathcal{D}$ or $(b, B') \notin \mathcal{D}$ for all $B' \in [b, A_b]$. But the latter case is impossible as $D \in [b, B] \subseteq [b, A_b]$ and $(b, D) \in \mathcal{D}$ (as $(b, C) \in \mathcal{D}$ and $\mathcal{D}$ is open). Hence $(b, A_b) \in \mathcal{D}$, and as $(b, B) \leq (b, A_b)$, $(b, B) \in \mathcal{D}$. Since D is arbitrary, (a, B) captures $\mathcal{D}$. □

Proposition 3.9. $\mathbb{M}_{\mathcal{R}}$ *has the Mathias property.*

Proof. Let A_G be an $\mathbb{M}_{\mathcal{R}}$-generic element of $\mathcal{R}$ (over some ground model $\mathbf{V}$), and let $D \leq A_G$. We need to show that the filter:

$$H := \{(a, A) \in \mathbb{M}_{\mathcal{R}} : D \in [a, A]\}$$

is generic. Let $\mathcal{D} \subseteq \mathbb{M}_{\mathcal{R}}$ be a dense open subset (in the ground model $\mathbf{V}$). For each $a \in \mathcal{AR}$, we let:

$$\mathcal{E}_a := \{B \in \mathcal{R} : a \in \mathcal{AR}{\restriction}B \text{ and } (a, B) \text{ captures } \mathcal{D}\}.$$

By Lemma 3.8, $\mathcal{E}_a \cap [a, A]$ is dense open in $[a, A]$ for all $A \in \mathcal{R}$ and $a \in \mathcal{AR}{\restriction}A$. Applying Lemma 2.5 to the family $\{\mathcal{E}_b \cap [b, A]\}_{b \in \mathcal{AR}{\restriction}A}$, we have that for all $(a, A) \in \mathbb{M}_{\mathcal{R}}$, there exists some $B \in [a, A]$ such that for all $b \in \mathcal{AR}{\restriction}B$ with $\mathrm{depth}_B(b) \geq \mathrm{lh}(a)$, there exists some $A_b \in \mathcal{E}_b$ such that $[b, B] \subseteq [b, A_b]$. That is, the set:

$$\begin{aligned}\mathcal{D}' := \{(a, A) \in \mathbb{M}_{\mathcal{R}} : \forall b \in \mathcal{AR}{\restriction}A\, (\mathrm{depth}_A(b) \geq \mathrm{lh}(a)\\ \to \exists A_b \in \mathcal{E}_b\, [b, A] \subseteq [b, A_b])\}\end{aligned}$$

is dense in $\mathbb{M}_{\mathcal{R}}$. Since G_{A_G} is generic, we may let $(a, A) \in G_{A_G} \cap \mathcal{D}'$, so $A_G \in [a, A]$ and $D \leq A$. Let n be such that $\mathrm{depth}_A(r_n(D)) \geq \mathrm{lh}(a)$, let $d := r_n(D)$, and let $A_d \in \mathcal{E}_d$ be such that $[d, A] \subseteq [d, A_d]$. Then $D \in [d, A] \subseteq [d, A_d]$, and (d, A_d) captures $\mathcal{D}$, so by the absoluteness of capturing (see the remark after Definition 3.7) there exists some initial segment $e \in \mathcal{AR}{\restriction}[d, D]$ of D such that $(e, A_d) \in \mathcal{D}$. Then $D \in [e, A_d]$, so $(e, A_d) \in H \cap \mathcal{D}$. Since $\mathcal{D}$ is arbitrary, H is a generic filter, as required. □

Proposition 3.10. $\mathbb{M}_{\mathcal{R}}$ *has the Prikry property.*

Proof. Let φ be a sentence in the forcing language. Let:

$$\mathcal{D} := \{q \in \mathbb{M}_{\mathcal{R}} : q \Vdash \varphi\}.$$

Note that $\mathcal{D}$ is open. We consider doing combinatorial forcing w.r.t. the set $\overline{\mathcal{D}}$. Let $(a, A) \in \mathbb{M}_{\mathcal{R}}$. By Lemma 3.4(5–6), there exists some $B \in [a, A]$ that decides a.

- Suppose that B accepts a, so $[a, B] \subseteq \overline{\mathcal{D}}$. We shall show that no $(b, C) \leq (a, B)$ forces $\neg\varphi$, so that $(a, B) \Vdash \varphi$. Otherwise, there exists some $(b, C) \leq (a, B)$ such that $(b, C) \Vdash \neg\varphi$, and we may assume that $b \sqsubseteq C$. Since $C \in [a, B] \subseteq \overline{\mathcal{D}}$, $C \in [a', B']$ for some $(a', B') \in \mathbb{M}_{\mathcal{R}}$ such that $(a', B') \Vdash \varphi$. Let $n \geq \max\{\mathrm{lh}(b), \mathrm{lh}(a')\}$, and we have that $(r_n(C), C) \leq (b, C)$ and $(r_n(C), C) \leq (a', B')$, so $(r_n(C), C)$ forces both φ and $\neg\varphi$, a contradiction. Thus $(a, B) \leq_0 (a, A)$ and $(a, B) \Vdash \varphi$.
- Suppose that B rejects a. By Lemma 3.5, replacing B by an element of $[a, B]$, we may assume that B rejects every $b \in \mathcal{AR}\restriction[a, B]$. We shall show that no $(b, C) \leq (a, B)$ forces φ, so that $(a, B) \Vdash \neg\varphi$. Otherwise, $(b, C) \in \mathcal{D}$ for some $(b, C) \leq (a, B)$, and we may assume that $a \sqsubseteq b \sqsubseteq C$. Then $[b, C] \subseteq \overline{\mathcal{D}}$, i.e. C accepts b. On the other hand, $b \in \mathcal{AR}\restriction[a, B]$ and $C \leq B$, so C rejects b by Lemma 3.4(3), which is impossible. Thus $(a, B) \leq_0 (a, A)$ and $(a, B) \Vdash \neg\varphi$.

□

4. Proof of Theorem 1.2

Throughout this section, we fix a closed triple $(\mathcal{R}, \leq, r)$ satisfying **A1**–**A4** such that $\mathcal{AR}$ is countable. Since $\mathcal{AR}$ is countable, we may assume that $\mathcal{AR} \subseteq \mathbf{V}_\omega$, so that $\mathcal{R} \subseteq \mathcal{AR}^{\mathbb{N}}$ is a set of reals, and lh, $\sqsubseteq$ and $\leq_{\mathrm{fin}}$ are subsets of $\mathbf{V}_\omega$ (hence are coded by reals).

For a transitive model M of set theory containing $\mathcal{AR}$, lh, $\sqsubseteq$ and $\leq_{\mathrm{fin}}$, we write $\mathcal{R}^M$ for $\mathcal{R}$ as computed in M. Note that closedness and the axioms **A1**–**A4** are absolute between such models with the same ordinals. This follows from Shoenfield absoluteness, as closedness and **A1**–**A4** are Π^1_2 statements (except for **A3**(2), which is Π^1_3 as written, but one can show that for $a \in \mathcal{AR}\restriction B'$, $[a, B'] \subseteq [a, B]$ iff $\mathcal{AR}\restriction[a, B'] \subseteq \mathcal{AR}\restriction B$, which is arithmetical).

The following lemma is due to Solovay, and a proof may be found in [4].

Lemma 4.1 (Solovay)**.** *Let κ be an inaccessible cardinal, let $\mathbb{P} \in \mathbf{V}$ be a forcing poset of size $< \kappa$, and let G be $\mathrm{Coll}(\omega, <\kappa)$-generic over $\mathbf{V}$.*

1. *For every $p \in \mathbb{P}$, there exists some $H \in \mathbf{V}[G]$ that is $\mathbb{P}$-generic over $\mathbf{V}$ such that $p \in H$.*
2. *If $H \in \mathbf{V}[G]$ is $\mathbb{P}$-generic over $\mathbf{V}$, then there exists some G^* that is $\mathrm{Coll}(\omega, <\kappa)$-generic over $\mathbf{V}[H]$ such that $\mathbf{V}[H][G^*] = \mathbf{V}[G]$.*

We are now ready to prove Theorem 1.2. The proof follows that of [6, Theorem 1.7].

Proof of Theorem 1.2. Let $\mathcal{X} \in \mathbf{L}(\mathbb{R})^{\mathbf{V}[G]}$ be a subset of $\mathcal{R}$. Note that $\mathcal{R}^{\mathbf{V}[G]} = \mathcal{R}^{\mathbf{L}(\mathbb{R})^{\mathbf{V}[G]}}$, as every element of $\mathcal{AR}^{\mathbb{N}} \cap \mathbf{V}[G]$ is coded by a real, so it suffices to show that $\mathcal{X}$ is Ramsey in $\mathbf{V}[G]$. Fix some $A \in \mathcal{R}^{\mathbf{V}[G]}$ and $a \in \mathcal{AR}\restriction A$. We need to show that there exists some $B \in [a, A]$ such that $[a, B] \subseteq \mathcal{X}$ or $[a, B] \subseteq \mathcal{X}^c$.

Since every set in $\mathbf{L}(\mathbb{R})^{\mathbf{V}[G]}$ is ordinal-definable from some real in $\mathbf{V}[G]$, there exist a formula ϕ, some real $r \in \mathbf{V}[G]$ and a finite sequence of ordinals α such that

for all $B \in \mathcal{R}^{\mathbf{V}[G]}$:

$$B \in \mathcal{X} \iff \mathbf{V}[G] \models \phi[B, r, \alpha].$$

Let $\phi^*(x, y, w)$ be the formula asserting that $\mathbb{1} \Vdash_{\mathrm{Coll}(\omega, <\kappa)} \phi(\check{x}, \check{y}, \check{w})$. By the κ-chain condition of $\mathrm{Coll}(\omega, <\kappa)$, there exists some $\lambda < \kappa$ such that $r, A \in \mathbf{V}[G\restriction\lambda]$. Let $\mathbf{W} := \mathbf{V}[G\restriction\lambda]$, and note that κ remains inaccessible in $\mathbf{W}$ and $\mathbf{V}[G] = \mathbf{W}[G\restriction[\lambda, \kappa)]$. We consider, in $\mathbf{W}$, the abstract Mathias forcing $\mathbb{M} := (\mathbb{M}_{\mathcal{R}})^{\mathbf{W}}$, and note that $(a, A) \in \mathbb{M}$. For $B \in \mathcal{R}^{\mathbf{V}[G]}$, recall from the beginning of §3 that $G_B := \{(c, C) \in \mathbb{M} : B \in [c, C]\}$, so that B is $\mathbb{M}$-generic over $\mathbf{W}$ iff G_B is an $\mathbb{M}$-generic filter over $\mathbf{W}$, in which case $B = A_{G_B}$ and $\mathbf{W}[G_B] = \mathbf{W}[B]$ (Lemma 3.2).

Claim 4.2. *Let $B \in \mathcal{R}^{\mathbf{V}[G]}$ be $\mathbb{M}$-generic over $\mathbf{W}$. Then:*

$$B \in \mathcal{X} \iff \mathbf{W}[B] \models \phi^*[B, r, \alpha].$$

Proof. Since $\mathbf{W} \models |\mathbb{M}| \leq 2^{\aleph_0} < \kappa$ and $G_B \in \mathbf{V}[G]$ is $\mathbb{M}$-generic over $\mathbf{W}$, by Lemma 4.1(2) applied in $\mathbf{W}$ there exists some $\mathrm{Coll}(\omega, <\kappa)$-generic G^* over $\mathbf{W}[B]$ such that $\mathbf{W}[B][G^*] = \mathbf{V}[G]$. Therefore:

$$\begin{aligned} B \in \mathcal{X} &\iff \mathbf{V}[G] \models \phi[B, r, \alpha] \\ &\iff \mathbf{W}[B][G^*] \models \phi[B, r, \alpha] \\ &\iff \mathbf{W}[B] \models \phi^*[B, r, \alpha], \end{aligned}$$

where the last equivalence follows from the homogeneity of $\mathrm{Coll}(\omega, <\kappa)$. ■

Let $\dot{A}$ be the canonical $\mathbb{M}$-name for the $\mathbb{M}$-generic element of $\mathcal{R}$. By the Prikry property of $\mathbb{M}$ (Proposition 3.10), there exists some $(a, A') \leq_0 (a, A)$ in $\mathbb{M}$ which decides the sentence $\phi^*(\dot{A}, \check{r}, \check{\alpha})$. By Lemma 4.1(1) applied in $\mathbf{W}$, there exists some $H \in \mathbf{V}[G]$ that is $\mathbb{M}$-generic over $\mathbf{W}$ such that $(a, A') \in H$. Let $A_H \in \mathcal{R}^{\mathbf{V}[G]}$ be the $\mathbb{M}$-generic element of $\mathcal{R}$ such that $H = G_{A_H}$ (which exists by Lemma 3.2), and note that $A_H \in [a, A'] \subseteq [a, A]$. We shall finish the proof by showing that $[a, A_H] \subseteq \mathcal{X}$ or $[a, A_H] \subseteq \mathcal{X}^c$.

Suppose that $(a, A') \Vdash \phi^*(\dot{A}, \check{r}, \check{\alpha})$, and let $B \in [a, A_H]$. By the Mathias property of $\mathbb{M}$ (Proposition 3.9), B is also $\mathbb{M}$-generic over $\mathbf{W}$. Since $B \leq A_H \leq A'$, we have $B \in [a, A']$, so $(a, A') \in G_B$, and as $\dot{A}^{G_B} = B$, $\mathbf{W}[B] \models \phi^*[B, r, \alpha]$. Hence $B \in \mathcal{X}$ by Claim 4.2. Therefore, $[a, A_H] \subseteq \mathcal{X}$. Alternatively, if $(a, A') \Vdash \neg\phi^*(\dot{A}, \check{r}, \check{\alpha})$, then $[a, A_H] \subseteq \mathcal{X}^c$. □

We remark that Theorem 1.2 appeared as [7, Theorem 6.1], with the following justification: Since every subset of a Polish space has the property of Baire in the Solovay model and the Ellentuck topology refines the Polish topology, every subset of $\mathcal{R}$ is Ramsey by the abstract Ellentuck theorem. This is incorrect, as a set with the property of Baire with respect to a topology need not have the property of Baire with respect to a finer topology. In fact, the implication "every metrically Baire subset of $[\mathbb{N}]^\infty$ is Ramsey" fails in ZFC: Let $A \in [\mathbb{N}]^\infty$ be coinfinite, and let $\mathcal{X} \subseteq [A]^\infty$ be any non-Ramsey set. Since $[A]^\infty$ is (metrically) nowhere dense, so is $\mathcal{X}$, so $\mathcal{X}$ is metrically Baire but not Ramsey.

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